

SGD Optimizer

Shuffle Dataset

Forward pass
(perceptron)

Compute error
(target - pred)

Clamp to $\{-1, 0, +1\}$

Update each weight
 $w \leftarrow w + lr * input * error$

Clamp to $\{-1, 0, +1\}$

Convert back to ternary

HillClimber

Pick 1-3 random
weights/biases

Set to random
 $\{0, 1, 2\}$

Evaluate accuracy
on full dataset

Better?
→ Keep mutation

Same?
→ Keep (explore)

Worse?
→ Revert

Restore best
if stuck